

Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
First Semester 2024-2025

Comprehensive Examination
(EC-3 Makeup)

Course No. : DSECLZG522
Course Title : Big Data Systems
Nature of Exam : Open Book
Weightage : 40%
Duration : 2.5 Hours
Date of Exam : 05/04/2025 (EN)

No. of Pages	= 3
No. of Questions	= 10

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Q.1. What are the factors contributing to MTBF? Define Availability. When do you say that a system is highly available? A middleware product has memory leaks and core dumps every 10 hours on an average. Upon failure, it is automatically restarted and will be back into operation within 3 minutes. Find out the availability of the middleware product?

[Marks 4]

Q.2. You have a file of size 928 MB stored on HDFS configured with Hadoop 2.x distribution. A data analytics program uses this file and runs in parallel across the cluster nodes.

(a) Default block size and replication factor is used in the configuration. What will be the total number of blocks including replicas stored in the cluster? What are the unique HDFS block sizes you will find for this specific file?

(b) The cluster has 64 cores to speed up the processing. If the program can at best achieve 60% parallelism in the code to exploit the multiple cores and the rest of it is sequential, what is the theoretical limit on speed-up you can expect with 64 cores compared to a sequential version of the same program running on one core with the same file? How will this limit change if you double the compute power to 128 cores? You can simplify the system to assume cluster nodes and cores mean the same and we can ignore the overheads of communication etc. depending on the specific cluster configuration, scheduling etc.

(c) Suppose you could use a more scalable algorithm with 80% parallelism and process larger files as you move to a 128-core system. What would be the theoretical speed-up limit for the 128 core system?

[Marks 4]

Q.3. What is meant by Tunable consistency in NoSQL databases? What is the effect of Write and Read latency on the consistency levels. When a NoSQL application is deployed on Single datacenter, what consistency levels are recommended by you for (a) Write intensive applications, and for (b) Read intensive applications?

[Marks 4]

- Q.4. An enterprise company need to address the following challenges with Big Data Analytics:
- Struggling with a Java based ELT tool to manage and extract value from the growing volume and variety of data and need to unify information across federated sources. Each disparate MySQL server records around 2 million updates, a total of 50 million updates daily.
 - Unable to relate raw data collected from system logs, sensors, or click streams with customer and line-of-business data managed in enterprise systems.
 - Each end-to-end ELT process consumes 14 hours. This process should be completed within an hour.
 - Some data warehousing queries on structured data are available only in SQL and there is no expertise available for transforming these SQL queries into Hadoop eco system tools.
- Recommend suitable solutions for the above challenges by making use of Hadoop Ecosystem products/tools. Give proper justifications for your recommendations. [Marks 4]

- Q.5. You are given access to 10 Terabyte of data with details of daily sales transactions belonging to an e-commerce company. The transactional data files are stored in a folder on HDFS with a structure as shown below:

Transaction No.	Invoice	Stock Code	Description	Quantity	Invoice Date	Price(\$)	Customer ID	Country
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It is not required to modify the data. You are not allowed to copy or move the data file into any other storage location. Your team has expertise in using SQL queries. Select a suitable analytics tool from the Hadoop ecosystem family to run queries on this data. Give justifications for your selection. Using the tool selected by you, write a query to find the total sales revenue received in \$ from each country. [Marks 4]

- Q.6. An HBase instance is configured to store 100000 rows in one Hfile. When all these Hfiles became full, compaction is initiated. Before compaction there were 10 Hfiles. Updated rows occupy 20% rows in each of these Hfiles. Assume that the updates happened only once. And 10% of the rows in each of these Hfiles were newly added. And 10% of the rows in each of the Hfiles were tagged as DELETED rows. Find out the total number of Hfiles resulting from the compaction done on the 10 Hfiles. [Marks 4]
- Q.7. You are tasked with building a real-time data processing pipeline using Apache Spark. The data arrives in JSON format from a Kafka topic, and you need to perform transformations and aggregations in real-time before storing the results in a NoSQL database like Cassandra. Describe the architecture and the Spark components you would use for this task. [Marks 4]
- Q.8. Examine the Spark scala code given below:

```
val lines = new Array[String](3)
lines(0) = "zero one reject"
lines(1) = "one two reject"
lines(2) = "two two zero"
val stringRDD=sc.parallelize(lines,2)
println("Number of Partitions: "+stringRDD.getNumPartitions)
val wordsRDD = stringRDD.flatMap(x => x.split(" "))
val filteredRDD = wordsRDD.filter(x => x.startsWith("o") || x.startsWith("t"))
val wcRDD = filteredRDD.map(word => (word, 1)).reduceByKey(_ + _)
wcRDD.collect().foreach(println)
```

(a) What are the outputs produced by this code?
(b) Describe the RDD lineage in the above code?

[Marks 4]

- Q.9. Why is windowing needed in Stream computing? Data from a smart device is transmitted over TCP/IP connection to a Spark streaming analytics system. Data from the device arrives at the rate of 1 event per second. The batching interval in the streaming context is set to 5 seconds. The stream analytics system uses a stateful algorithm with an analysis window of 60 seconds. It is required to generate an output from the streaming system every 30 seconds. What type of windowing is recommended by you? Give appropriate configuration parameters for the Window suggested by you. What will be the number of RDDs in the Dstreams generated in the window duration. [Marks 4]
- Q.10. An eCommerce company is planning to launch a new application for online sale of items. During normal seasons of the year, this application deployed on a cluster of 4 servers will guarantee acceptable response to the customers. During festival seasons, the sale is expected to double. Which feature of cloud will be useful in minimizing the data center costs without compromising the response time. What are the recommended features of the components to be used in the design of this application for deployment on cloud? [Marks 4]
